

EventVault

AI-Powered Event & Media Management Platform

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The Problem: Media Chaos

Lost in the Noise

Event spaces, clubs, and production houses fight consistent operational friction:

- ✗ **Scattered Media:** Photos and videos scattered randomly across drive links, personal devices, and local chat groups.
- ✗ **Poor Searchability:** Zero asset context. Finding one specific image inside huge photo sets requires slow manual browsing.
- ✗ **No Access Control:** High-resolution originals are shared completely unprotected, without download tracking or viewer roles.
- ✗ **The Isolation Issue:** Attendees spend excessive time manually hunting through massive public galleries just to find a single photo of themselves.



Introducing EventVault

A unified, user-focused ecosystem handling complete media distribution.



Event Management

Build custom workspaces for clubs, design dedicated event buckets, and partition collections securely into albums.



AI Discovery

Leverage computer vision to scan media arrays, extracting instant smart tags and coordinating biometrics.



Secure Sharing

Enforce file protection with resilient AWS S3 object storage paths and dynamic text watermarking on download.

EventVault Roles & Permissions



Super Admin

- ✓ Full Platform Control
- ✓ Manage Clubs & Users
- ✓ Delete Any Content



Photographer

- ✓ Upload Media
- ✓ AI Suggestions
- ✓ Manage Albums



Club Member

- ✓ Join Clubs
- ✓ Upload Event Media
- ✓ Access Private Events



Viewer

- ✓ View Public Content
- ✓ Like & Comment
- ✓ Bookmark Media
- ✓ No Upload Permissions



Club Hierarchy



OWNER



ADMIN



MEMBER

Hybrid access control combining platform roles and club-level permissions.

The Feature Ecosystem



Club & Organizer Spaces

Create distinct clubs, handle member join requests, drop profiles, and configure multi-level event privacy scopes.



High-Volume Bulk Ingestion

Drag-and-drop huge asset libraries smoothly. Background workers queue processes instantly for quick parallel transfers.



Social Engagement Layers

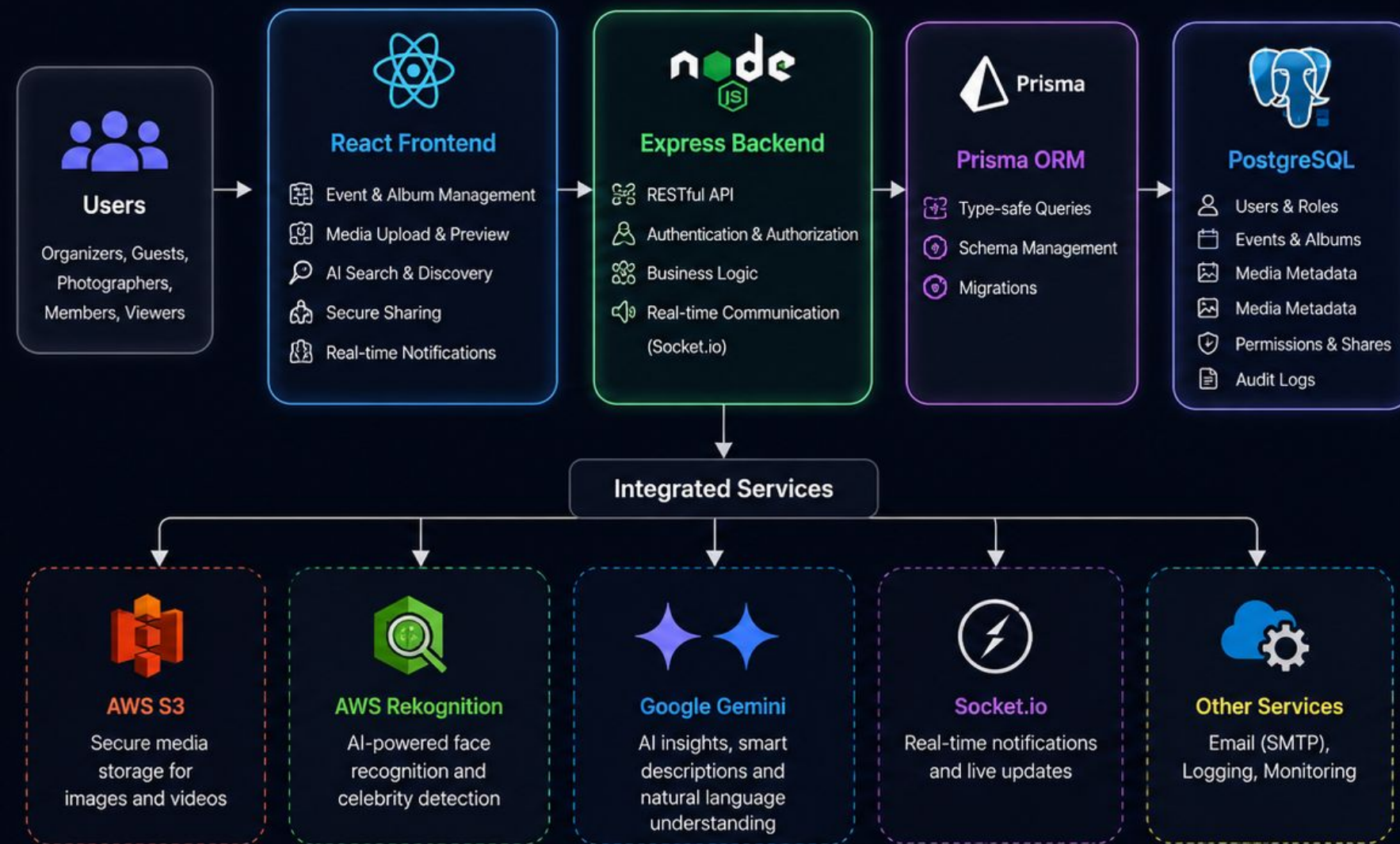
Drive real interaction. Drop likes, append comments, save target media to favorites, and browse clean contextual logs.



Personalized User Galleries

Attendees upload a single reference selfie to unlock a tailored view displaying every match found across all photo sets.

System Architecture



Production-Ready Core Stack

- React Frontend:** Fast client workspace wrapped in TypeScript.
- Express Backend:** Robust API router controlling business logic.
- Prisma ORM & PostgreSQL:** Type-safe relational storage layer.

Integrated Services

AWS S3 (Storage)

AWS Rekognition (AI)

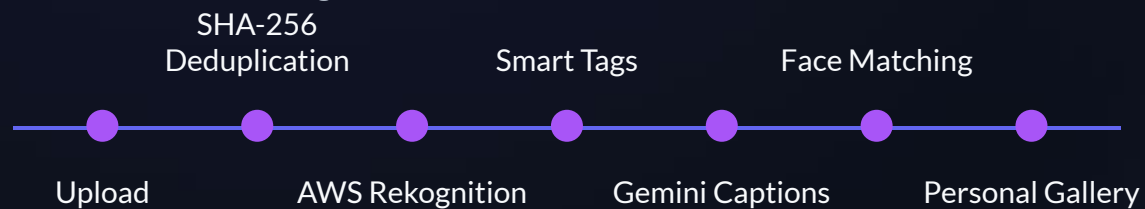
Google Gemini (Caption)

Socket.io (Realtime Bus)

Intelligence Pipeline

Automated Asset Insight

Every upload triggers an immediate multi-stage processing pipeline that turns raw images into searchable database records:



Asset Safety & Governance



Cryptographic User Isolation

Secure JWT sessions with rolling tokens enforce permissions down to the route level. Restricted viewer states keep public spaces strictly read-only.

Dynamic Contextual Watermarking

Egress paths intercept downloads on the fly. Server buffers overlay custom stamps detailing the **Club Name**, **Event Identity**, and **Downloader Role**.

Protected Cloud Repository

Raw image blocks stay isolated in locked S3 environments. Media items stream directly to authorized clients via brief, expiring AWS Presigned URLs.

Built for Scale



AWS S3 Storage

Highly durable cloud media storage handles extensive asset loads with minimal ingestion delays.



Real-Time Notifications

Socket.io streams deliver immediate match metrics and interaction updates directly to users without constant DB polling.



Turborepo Monorepo

Shared infrastructure components and isolated compilation targets allow stable development and code reuse.



PostgreSQL + Prisma

Type-safe relational database queries keep transactional data intact, even under high traffic loads.

The Road Ahead



Global Edge CDN

Deploy multi-region media caches to lower download latencies globally.



Mobile PWA Support

Add installable app wrappers with service workers for offline asset caching.



QR Code Album Sharing

Generate instant codes for fast, proximity-based onboarding at physical venues.



Analytics Telemetry

Provide clubs with user activity monitoring and storage growth tracking dashboard views.



Collaborative Albums

Allow several verified contributors to drop photos simultaneously into one timeline.

Thank You!
